

Chapter 6: AI Agents

Learning Objectives

By the end of this chapter, you will:

- Understand what AI Agents are and how they differ from traditional chatbots and LLM applications.
- Learn the concepts of tools, memory, planning, and multi-agent systems.
- Understand how AI TPMs lead enterprise AI Agent initiatives from ideation to production.

Beginner Explanation

An AI Agent is an AI system that can:

- Understand goals.
- Make decisions.
- Use tools.
- Execute actions.
- Remember context.
- Complete tasks with minimal human intervention.

Think of the evolution like this:

Technology	Capability
Search Engine	Finds information
Chatbot	Answers questions
LLM	Generates content
RAG System	Uses enterprise knowledge
AI Agent	Performs actions

Example

Imagine asking:

"Schedule a meeting with the Finance team next week, summarize the agenda, and email attendees."

A chatbot might respond with instructions.

An AI Agent could:

1. Access your calendar.
2. Identify available slots.

- 3. Create the meeting invite.
- 4. Draft the agenda.
- 5. Send emails.
- 6. Notify stakeholders.

This ability to perform actions is what makes AI Agents powerful.

Core Components of an AI Agent

Component	Purpose
LLM	Reasoning engine
Tools	Execute actions
Memory	Store context
Planner	Determine next steps
Retriever	Access enterprise data
Guardrails	Ensure safe execution

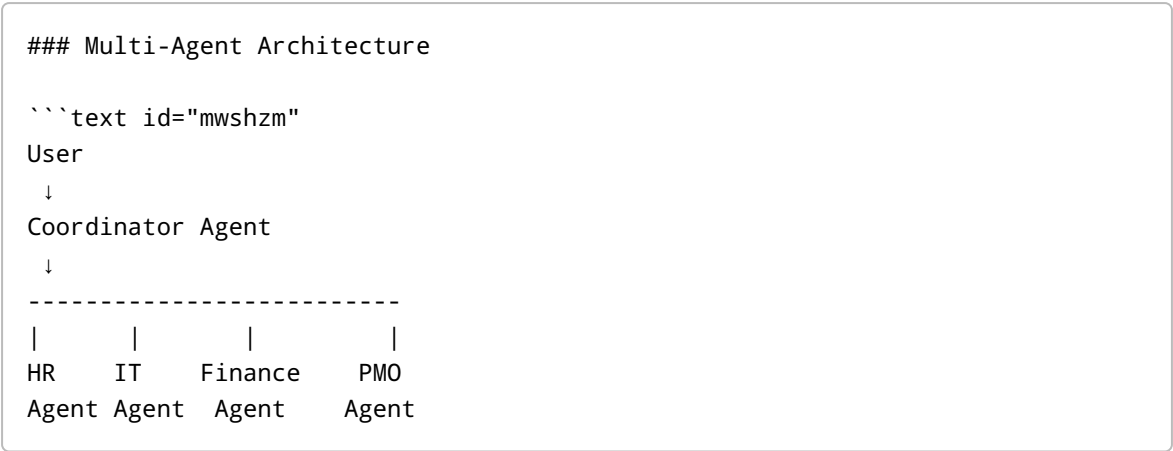
Intermediate Explanation

Enterprise AI Agents are typically built using four capabilities:

- 1. Planning
- 2. Reasoning
- 3. Tool Usage
- 4. Memory

Single-Agent Architecture

``text id="uivob6" User ↓ Agent ↓ Tools ↓ Response



Common Enterprise Agent Types

Agent Type	Function
Helpdesk Agent	Resolve IT incidents
HR Agent	Employee support
TPM Agent	Program dashboards and reports
Finance Agent	Expense analysis
Security Agent	Threat monitoring
Procurement Agent	Vendor management

Enterprise Challenges

- Access control
- Cost management
- Agent coordination
- Compliance requirements
- Reliability concerns
- Observability limitations

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Coordinates development and deployment of AI Agents across teams.
Risk	Addresses security, compliance, and autonomous decision-making risks.
Stakeholder Management	Aligns expectations regarding agent capabilities and limitations.
Budget	Tracks infrastructure, API, and operational costs.

Enterprise Example

Business Problem

A large technology company managed more than 300 active programs globally. TPMs spent considerable time creating:

- Weekly reports
- RAID logs
- Executive summaries
- Resource dashboards

Solution

Build a TPM Copilot Agent capable of:

- Retrieving project data.
- Generating reports.
- Identifying risks.
- Escalating blockers.
- Drafting executive updates.

Architecture

- Azure OpenAI
- Azure AI Search
- Azure Functions
- Azure DevOps APIs
- Power BI
- Microsoft Teams

Outcome

Metric	Result
Time Saved	12 Hours per TPM Weekly
Report Accuracy	94%
Adoption Rate	85%
Annual Savings	~\$750,000

Lessons Learned

- Agents require strict governance.
- Human approval should remain for critical actions.
- Multi-agent systems improve scalability.
- Monitoring and auditability are mandatory.

Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Agent reasoning and orchestration
Azure AI Search	Enterprise knowledge retrieval
Azure Functions	Tool execution
Azure AI Foundry	Agent development and evaluations

Implementation Workflow

Step 1: Define Agent Goals

Examples:

- Schedule meetings.
 - Generate reports.
 - Analyze risks.
 - Summarize incidents.
-

Step 2: Configure Tools

Typical enterprise tools:

Tool	Function
Jira	Project updates
Azure DevOps	Sprint tracking
Outlook	Email actions
Teams	Notifications
ServiceNow	Incident management
Power BI	Dashboard retrieval

Step 3: Add Memory

Types of memory:

Memory Type	Purpose
Short-Term	Session context
Long-Term	User preferences
Shared	Team knowledge
Episodic	Historical interactions

Step 4: Implement Guardrails

Examples:

- Human approval workflows
- Access restrictions
- Rate limiting

- Audit logging
- Policy enforcement

Step 5: Monitor Performance

Track:

- Task completion rate
- Cost per action
- Latency
- User satisfaction
- Error rate

Mermaid Diagram

```
```mermaid id="av7k7f" graph TD
```

```
A[User Request]
```

```
A --> B[AI Agent]
```

```
B --> C[Azure OpenAI]
```

```
C --> D[Azure AI Search]
```

```
D --> E[Azure Functions]
```

```
E --> F[Enterprise Tools]
```

```
F --> G[Generated Actions]
```

```
G --> H[End User]
```

```

```

```
Multi-Agent Architecture
```

```
```mermaid id="pv9u7q"
graph TD
```

```
A[User]
```

```
A --> B[Coordinator Agent]
```

```
B --> C[HR Agent]
```

```
B --> D[Finance Agent]
```

B --> E[TPM Agent]

B --> F[IT Agent]

C --> G[Enterprise Systems]

D --> G

E --> G

F --> G

TPM Checklist

- ☐ Define agent objectives.
- ☐ Identify required tools.
- ☐ Establish approval workflows.
- ☐ Configure access controls.
- ☐ Define KPIs.
- ☐ Estimate operational costs.
- ☐ Implement monitoring.
- ☐ Conduct security reviews.
- ☐ Define escalation procedures.
- ☐ Obtain stakeholder approval.

Templates

Agent Readiness Assessment

Question	Status
Are agent goals defined?	Yes/No
Are tools identified?	Yes/No
Is governance approved?	Yes/No
Are KPIs established?	Yes/No
Is monitoring configured?	Yes/No

Agent KPI Dashboard

Metric	Target
Task Completion Rate	>90%

Metric	Target
User Satisfaction	>90%
Latency	<5 Seconds
Cost per Request	<\$0.10
Availability	99.9%

Risk Register

Risk	Impact	Mitigation
Unauthorized Actions	High	Human approval
Cost Overruns	Medium	Budget alerts
Data Leakage	High	RBAC
Tool Failure	Medium	Retry mechanisms

Common Mistakes

1. Giving agents excessive permissions.
2. Deploying agents without governance.
3. Assuming agents are fully autonomous.
4. Ignoring observability requirements.
5. Underestimating operational costs.
6. Failing to implement human-in-the-loop processes.
7. Skipping security assessments.

Key Takeaways

- AI Agents extend LLMs by adding tools, memory, and autonomous actions.
- Enterprise agents should be governed, monitored, and auditable.
- Multi-agent architectures improve scalability and specialization.
- Human approval remains essential for critical business processes.
- Azure provides a comprehensive platform for building enterprise AI Agents.
- AI TPMs play a critical role in balancing innovation, risk, cost, and business value.

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